

AIRBUS GLOBAL QUANTUM + AI CHALLENGE 2026

Quantum Aero Solver

A block-encoded Carleman-LBM concept for the convecting Taylor-Green vortex

CONCEPT PROPOSAL

TRACK A | FTQC / LBM

Prepared by: Mohaimin | Team Quantum Aero Solver

Date: 4 September 2026

Status: validated concept; quantum advantage not yet claimed

1. Executive summary

Airbus seeks a more scalable route to high-fidelity aerodynamic simulation at conditions where classical computation and physical testing become costly. We propose a fault-tolerant quantum Lattice Boltzmann solver for the two-dimensional convecting Taylor-Green vortex. The concept combines a D2Q9 finite-velocity representation, Carleman linearization of the nonlinear BGK collision, a block encoding of the resulting non-unitary map, and exact unitary streaming on a periodic grid.

The proposal is deliberately end-to-end: advantage will be assessed at fixed physical accuracy and will include state preparation, time stepping, block-encoding normalization, amplitude amplification, observable extraction, routing, and fault-tolerant overhead. The target output is not the full velocity field; it is a small set of engineering observables such as kinetic-energy decay and selected Fourier modes, avoiding the data-extraction bottleneck that can erase a quantum speedup.

Core hypothesis. For high Reynolds number and selected low-dimensional observables, a structured implementation of collision plus streaming can reach a lower asymptotic or crossover time-to-solution than a grid-converged classical reference at the same error tolerance. The project includes an explicit no-go gate if end-to-end resource accounting does not support that hypothesis.

90

lifted collision dimension

8

padded local logical qubits

1.77%

raw post-selection

Proposed contribution

- A corrected, periodic, analytically verifiable TGV benchmark with complete provenance.
- A physically meaningful nine-population collision map rather than a toy direction-bit gate.
- A structured block-encoding research path benchmarked against a dense upper bound.
- A transparent advantage claim based on selected observables and total cost, not qubit count alone.

2. Benchmark and physical model

The benchmark is the incompressible convecting Taylor-Green vortex on a periodic two-dimensional domain. It is challenging enough to expose nonlinear advection, viscous dissipation, numerical dispersion, and stability limits, while retaining an exact solution for rigorous fidelity measurement [1].

$$\nabla \cdot \mathbf{u} = 0, \quad \partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\rho^{-1} \nabla p + \nu \nabla^2 \mathbf{u}$$

We separate the periodic box length from the vortex length scale:

$$u = U_c + V_0 \sin\left(\frac{x - U_c t}{L_c}\right) \cos\left(\frac{y - V_c t}{L_c}\right) e^{-2\nu t/L_c^2}$$

$$v = V_c - V_0 \cos\left(\frac{x - U_c t}{L_c}\right) \sin\left(\frac{y - V_c t}{L_c}\right) e^{-2\nu t/L_c^2}$$

$$p = p_0 + \frac{\rho V_0^2}{4} \left[\cos\left(\frac{2(x - U_c t)}{L_c}\right) + \cos\left(\frac{2(y - V_c t)}{L_c}\right) \right] e^{-4\nu t/L_c^2}$$

Parameter	Value	Interpretation
Periodic domain	$[0, 2\pi) \times [0, 2\pi)$	$L_{\text{box}} = 2\pi$
Vortex scale	$L_c = 1$	Makes the exact field periodic
Velocity	$V_0 = 1, U_c = 1, V_c = 0$	Unit vortex in streamwise flow
Density / pressure	$\rho = 1, p_0 = 0$	Constant-density benchmark
Viscosity	$\nu = V_0 L_c / \text{Re}$	$\text{Re} = 10$ through $5000+$
Primary final time	$t = 1$	Matches the challenge illustration

Benchmark clarification. The challenge statement uses the same symbol L for a 2π domain and for the trigonometric vortex scale. Taken literally, that field is not periodic. We use $L_{\text{box}} = 2\pi$ and $L_c = 1$, state the convention in every artifact, and will request organizer confirmation before final scoring.

3. Numerical foundation: D2Q9 LBM

The classical control and quantum formulation share the same D2Q9 discretization. This removes discretization mismatch from the first quantum-versus-classical comparison. At each cell, nine populations encode density and momentum:

$$\begin{aligned}\rho &= \sum_{i=0}^8 f_i, & \rho \mathbf{u} &= \sum_{i=0}^8 f_i \mathbf{e}_i, & c_s^2 &= \frac{1}{3} \\ f_i^{eq} &= w_i \rho \left[1 + \frac{\mathbf{e}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u} \cdot \mathbf{u}}{2c_s^2} \right] \\ f_i^* &= f_i - \omega(f_i - f_i^{eq}), & f_i(\mathbf{x} + \mathbf{e}_i, t + \Delta t) &= f_i^*(\mathbf{x}, t), & \omega &= \tau^{-1}\end{aligned}$$

Validation metrics

$$\mathcal{E}_{L2} = \frac{\|\mathbf{u}_{sim} - \mathbf{u}_{exact}\|_2}{\|\mathbf{u}_{exact}\|_2}, \quad K(t) = \frac{1}{2} \langle u^2 + v^2 \rangle$$

- Total-field and vortex-only velocity L2 error, to prevent the uniform stream from hiding vortex error.
- Kinetic-energy decay, mass drift, momentum drift, and divergence norm.
- Grid, Mach, and timestep convergence before any advantage comparison.

Classical comparators

The in-repository BGK solver is the controlled algorithmic baseline, not the state-of-the-art comparator. The final study will add a Fourier pseudo-spectral or high-order finite-volume reference, use identical hardware-reporting rules, and compare time-to-solution at fixed error rather than at fixed grid size.

4. Proposed fault-tolerant quantum algorithm

1 Prepare Encode populations and selected observables	2 Collide Carleman lift + block encoding	3 Stream Unitary periodic D2Q9 permutation	4 Measure Energy, modes, error indicators
---	--	--	---

4.1 Carleman collision

Around reference density ρ_0 , the quadratic equilibrium is written as a linear and quadratic function of the nine populations. Define the order-2 lifted state and collision matrix:

$$f^{eq} \approx Lf + Q:(f \otimes f), \quad z = (f, f \otimes f)^T, \quad z^* = Mz$$

$$f^* = Rf + \omega Q:(f \otimes f), \quad (f \otimes f)^* \approx (R \otimes R)(f \otimes f), \quad R = (1 - \omega)I + \omega L$$

For one cell, z has dimension $9 + 81 = 90$. This is padded to 128 amplitudes. The non-unitary collision is normalized by $\alpha \geq \|M\|_2$ and embedded into a unitary:

$$A = M/\alpha, \quad U_A^{00} = A, \quad U_A^{01} = \sqrt{I - AA^\dagger}, \quad U_A^{10} = \sqrt{I - A^\dagger A}, \quad U_A^{11} = -A^\dagger$$

Post-selecting the signal ancilla applies A . Obvious amplitude amplification is included in the cost model; success probability is never treated as free.

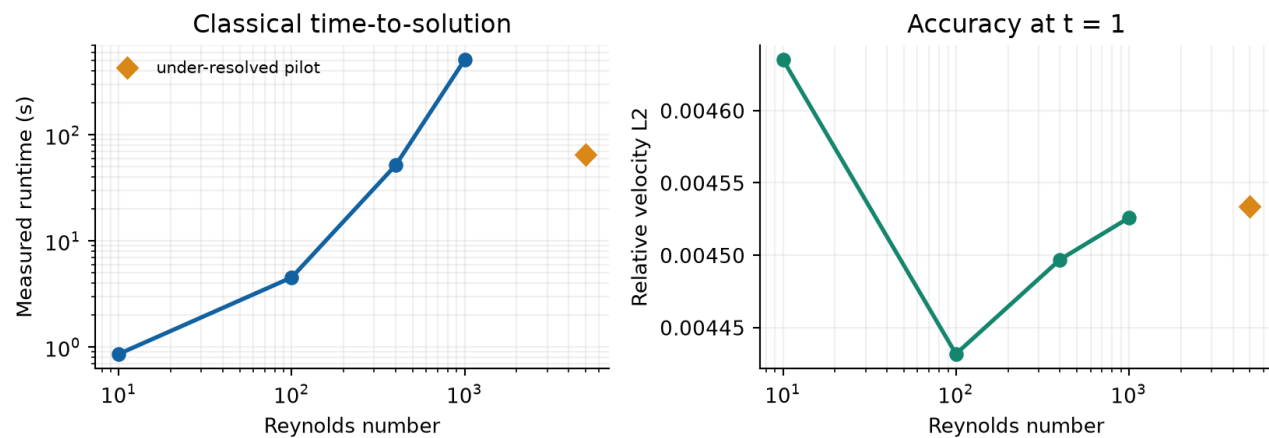
4.2 Streaming and observables

Streaming is a reversible, direction-controlled modular shift over the position register. The primary outputs are mean kinetic energy and selected Fourier coefficients. Full-field tomography is excluded from the claimed advantage path because its extraction cost scales with the field dimension.

Research gate: the present spatial validator rebuilds f tensor f classically. Milestone 2 must replace that step with a coherent global/local lift or adopt an end-to-end incompressible LBM construction with bounded advantage [4]. Failure triggers a no-go result, not an unsupported speedup claim.

5. Preliminary evidence and technical readiness

The corrected repository provides executable evidence for each local building block and records limitations alongside results. These figures are feasibility evidence, not a quantum-advantage claim.



Experiment	Measured result	Interpretation
Canonical datasets	Final L2 = 0.00443 to 0.00463	Legacy 25-33% error removed
Collision block	max error = 6.94e-17	90D map reproduced numerically
Dilation unitarity	max error = 5.11e-15	8-qubit dilation verified
Post-selection	p = 0.0177	56.4 raw attempts expected
Spatial fidelity	velocity L2 = 1.28e-5	Carleman versus exact BGK, N=32, t=0.1
Applied noise	Hellinger fidelity = 0.964	20,000 shots; noise actually passed to Aer
Routed dense synthesis	depth 107,128; CX 87,216	Upper-bound baseline; structure is essential
Re=5000 pilot	N=128, t=1, 65.5 s	Measured but explicitly under-resolved

The dense routed collision requires 51,483 one-qubit u gates and 87,216 CX gates on a nearest-neighbor ring. This is not the proposed production circuit; it is a measured upper bound that establishes the optimization target. A structured implementation must exploit D2Q9 sparsity, repeated coefficients, symmetry, and local collision reuse.

6. Experimental program

Every comparison uses the same physical domain, final time, Reynolds number, observable, and error tolerance. Each timing point receives warm-up runs and at least five measured repetitions for the final submission.

Work package	Sweep	Recorded outputs	Pass criterion
A. Physics convergence	Re 10, 100, 400, 1000, 2000, 5000; N 32-2048; Ma 0.1-0.0125	L2, energy, phase, mass, divergence	Grid and Mach convergence demonstrated
B. Collision fidelity	Carleman order 1-3; 1-1000 steps; density/velocity perturbations	Population and observable drift	Error budget below target epsilon
C. Quantum correctness	1, 2, 5, 10, 50 timesteps	State/observable error versus BGK	No hidden classical operation in claimed path
D. Noise sensitivity	Gate error 1e-4 to 1e-2; shots 1e3 to 1e6	Fidelity, bias, confidence interval	Noise model applied after transpilation
E. Resource scaling	N 8-2048; Re 10-5000; epsilon 1e-1 to 1e-4	Logical/physical qubits, T count, depth, calls	Complete cost ledger
F. Classical reference	BGK plus spectral/high-order FV	Runtime, peak memory, accuracy	Same tolerance and reporting protocol

Required figures for the final submission

- Time-to-solution versus Reynolds number at fixed error, with confidence intervals.
- Memory or physical-qubit requirement versus Reynolds number.
- Velocity L2 and kinetic-energy error versus Reynolds number and resolution.
- Total quantum cost broken into preparation, collision, streaming, amplification, and measurement.
- Crossover or no-crossover plot against both classical comparators.

Reproducibility

All configurations, seeds, package versions, CPU/GPU/backend specifications, transpiler settings, raw counts, and derived metrics are machine-readable. Failed and null experiments remain in the evidence trail.

7. Definition and test of quantum advantage

A smaller amplitude register is not, by itself, an advantage. The claim is accepted only if total fault-tolerant time or memory is lower at the same physical error for a declared observable. We therefore use the following cost ledger:

$$T_Q = T_{prep} + n_t[N_{BE}C_{AA}(p) + N_S]t_{cycle} + N_{shots}T_{meas}$$

Here N_{BE} is the block-encoding cost, $C_{AA}(p)$ captures post-selection or amplitude amplification, N_S is streaming cost, and N_{shots} is set by the observable precision. Rotation synthesis, routing, magic-state factories, logical cycle time, and failure probability are included in t_{cycle} for each hardware scenario.

Decision gate	Evidence required	Action
G1 - physical fidelity	Converged classical TGV and stable Carleman error	Proceed only if error budget closes
G2 - coherent timestep	No classical re-lifting inside claimed evolution	Proceed to FT resource compilation
G3 - structured collision	Material reduction from dense 107,128 depth	Reject dense implementation if no structure found
G4 - end-to-end crossover	$T_Q < T_C$ or $M_Q < M_C$ at fixed epsilon	Claim bounded advantage, otherwise report no-go

Primary claim boundary. The proposal targets bounded advantage for selected observables at useful error tolerance. It does not promise exponential advantage for full-field recovery. Recent end-to-end analysis identifies convergence, condition number, time stepping, and data extraction as decisive bottlenecks [4]; our gates test each one.

Hardware scenarios

- Logical all-to-all: algorithmic scaling and lower-bound comparison.
- Routed logical: nearest-neighbor connectivity and native basis decomposition.
- Fault tolerant: surface-code distance, factory throughput, physical qubits, and wall time.

8. Delivery plan, risks, and impact

Phase	Duration	Deliverable	Exit evidence
1. Benchmark lock	Weeks 1-2	Canonical equations, data, spectral/FV comparator	Convergence and provenance report
2. Coherent algorithm	Weeks 3-6	Global/local lifted timestep without classical re-lift	Multi-step state/observable fidelity
3. Circuit engineering	Weeks 7-10	Sparse structured block encoding and streaming	Native-basis and routed resources
4. Scale study	Weeks 11-13	Re and epsilon sweeps	Runtime, memory, error plots
5. FT assessment	Weeks 14-15	Physical resource scenarios	Crossover or no-go conclusion
6. Submission	Week 16	Technical report, code, datasets, reproducibility bundle	Independent rerun checklist

Top risks and mitigations

Risk	Likelihood / impact	Mitigation
Carleman truncation fails at high Re	High / high	Adaptive order, rescaling, stability map; pivot to bounded-advantage incompressible LBM
Post-selection cost dominates	High / high	Normalization optimization, amplitude amplification, local encodings; explicit stop gate
Dense synthesis remains exponential	High / high	Exploit D2Q9 sparsity and symmetry; quantify structure before scale claims
Data extraction removes speedup	High / high	Restrict outputs to energy and selected modes; compare full observable cost
Classical reference is too weak	Medium / high	Add spectral/high-order FV comparator and fixed-error methodology
PDF parameter ambiguity	Medium / medium	Separate L_box and L_c; request organizer confirmation

Expected impact

The project will deliver either a defensible bounded quantum advantage for selected aerodynamic observables or a quantified no-go boundary showing exactly which overhead prevents it. Both outcomes reduce uncertainty for future Airbus quantum-CFD investment. The validated benchmark, resource ledger, and reproducible datasets remain useful beyond the challenge for solver selection and hardware-roadmap studies.

9. References and evidence artifacts

- [1] Airbus. Quantum Solvers: Enhancing Predictive Aerodynamic Modeling Capabilities. Global Quantum + AI Challenge 2026 Enterprise Challenge Statement, 2026.
- [2] Y. H. Qian, D. d'Humieres, and P. Lallemand. Lattice BGK Models for Navier-Stokes Equation. Europhysics Letters 17(6), 1992.
- [3] J.-P. Liu et al. Efficient quantum algorithm for dissipative nonlinear differential equations. PNAS 118, e2026805118, 2021; corrected arXiv version 2026. <https://arxiv.org/abs/2011.03185>
- [4] D. Jennings et al. An end-to-end quantum algorithm for nonlinear fluid dynamics with bounded quantum advantage. arXiv:2512.03758, 2025. <https://arxiv.org/abs/2512.03758>
- [5] A. D. Bastida Zamora et al. Quantum lattice Boltzmann method for several time steps: A local Carleman linearization algorithm. arXiv:2511.13072v3, 2026. <https://arxiv.org/abs/2511.13072>
- [6] A. Gilyen, Y. Su, G. H. Low, and N. Wiebe. Quantum singular value transformation and beyond. STOC 2019. <https://arxiv.org/abs/1806.01838>
- [7] A. M. Childs, R. Kothari, and R. D. Somma. Quantum algorithm for systems of linear equations with exponentially improved dependence on precision. SIAM Journal on Computing 46(6), 2017.

Repository evidence

Artifact	Purpose
quantum_aero/classical.py	Corrected periodic D2Q9 solver and diagnostics
quantum_aero/carleman.py	Nine-population Carleman map and unitary dilation
quantum_aero/quantum.py	Applied-noise experiment and routed resource counts
results/validation.json	Machine-readable block, noise, fidelity, and resource evidence
results/reynolds_sweep.csv	Measured $t=1$ Reynolds sweep including under-resolved $Re=5000$ pilot
baseline/dataset/*.h5	Corrected 21-snapshot datasets with embedded provenance

Submission statement. Preliminary results establish correctness of the local collision block and benchmark pipeline. They do not yet establish an end-to-end quantum advantage. The experimental gates in this proposal define the evidence required before such a claim will be made.